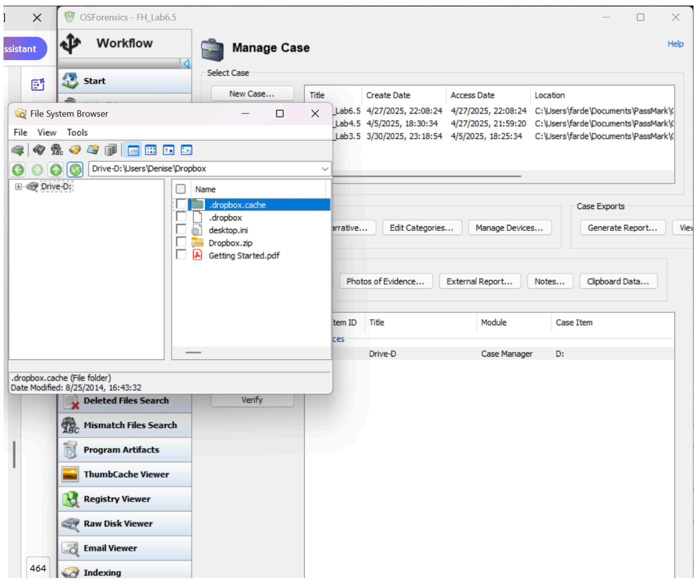
CPEG495 Lab 6.5: Hands On Projects

By Furdeen Hasan

Hands-On Project 13-1:

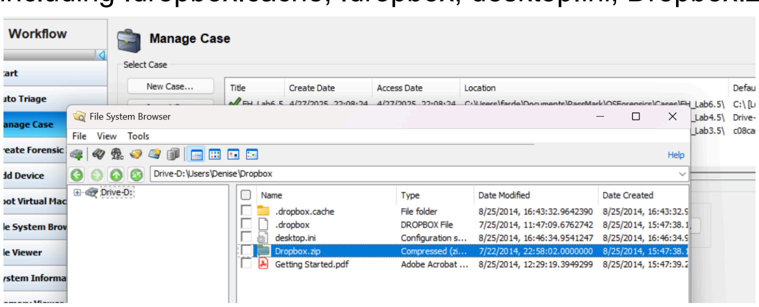
1. Opening the Case in OSForensics

Loaded the forensic image InCh05 and created a new case named FH_Lab6.5 in OSForensics to begin evidence review.

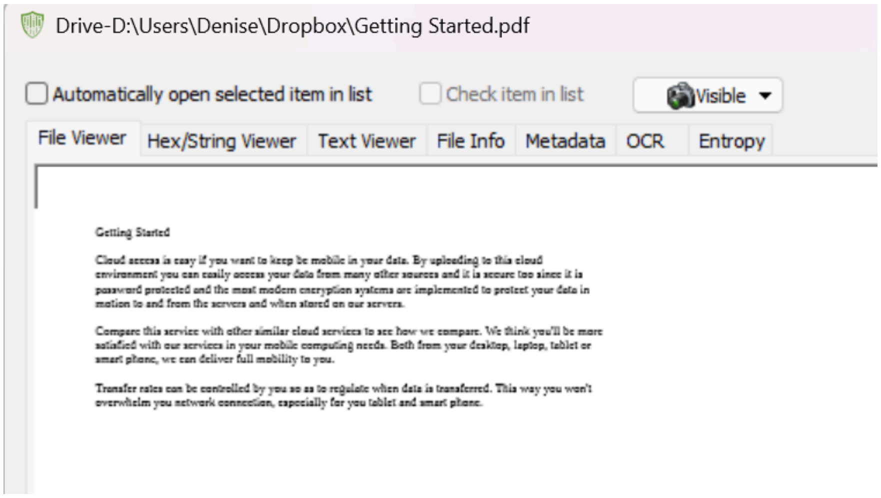


2. Exploring Dropbox Directory

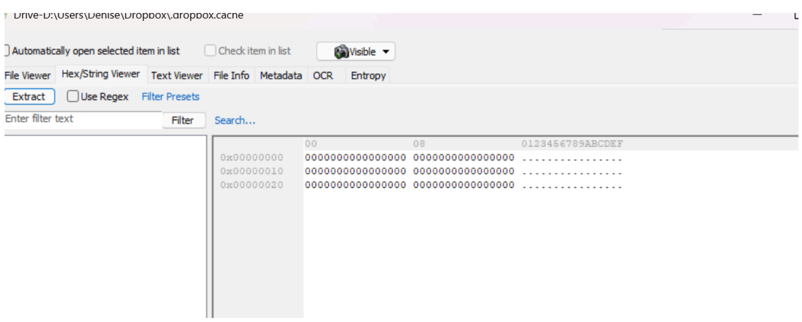
Navigated to Drive D:\Users\Denise\Dropbox where several files and folders were identified, including .dropbox.cache, .dropbox, desktop.ini, Dropbox.zip, and Getting Started.pdf.



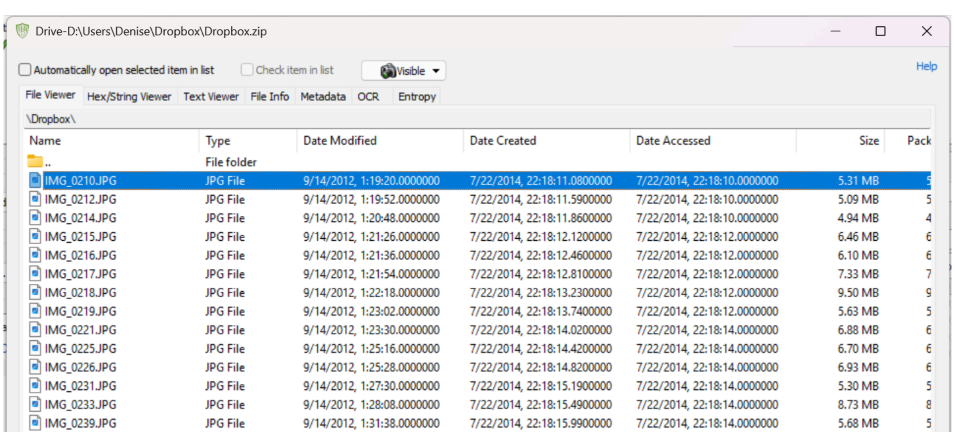
3. Viewing 'Getting Started.pdf'
Opened and previewed the Getting Started.pdf file, which discusses cloud service usage but did not appear immediately suspicious.



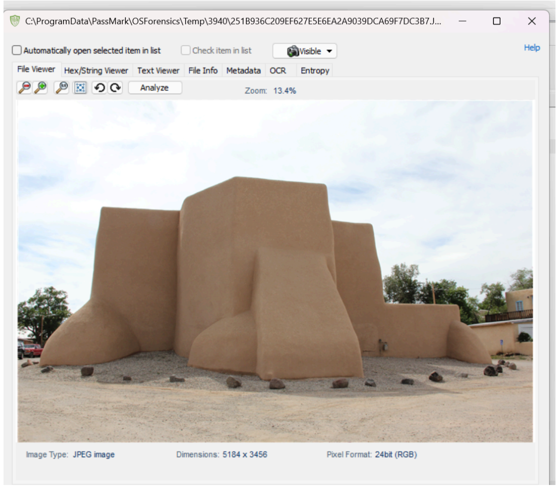
4. Analyzing .dropbox.cache
The .dropbox.cache folder contained no usable content—hex view confirmed it was filled with zeroed-out data.



5. Identifying Dropbox.zip File
Note the presence of a large compressed file Dropbox.zip, modified and created on 7/22/2014. It was flagged for further examination. Extracted and analyzed the contents of Dropbox.zip, revealing multiple JPG and PNG images related to architectural and scenic locations.

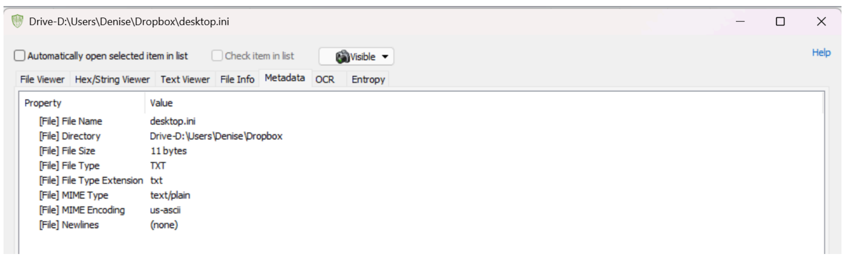


6. Image found in Dropbox.zip: Sampled a few images inside the archive. The images appeared to show photographs of adobe-style buildings and churches.



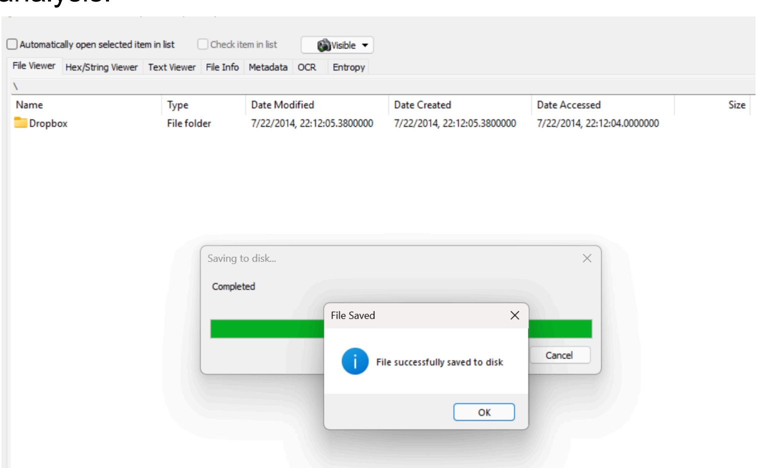
7. Investigating 'desktop.ini' Metadata

Checked the desktop.ini file and confirmed it was a standard configuration file (size: 11 bytes, type: TXT), likely created automatically by Windows.

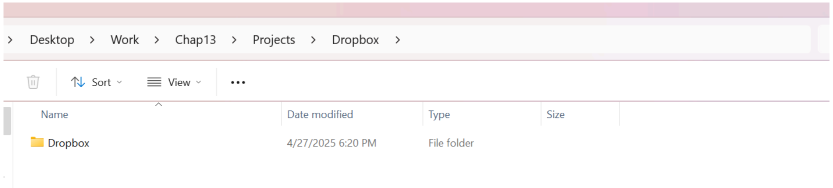


8. Saving the Dropbox Folder

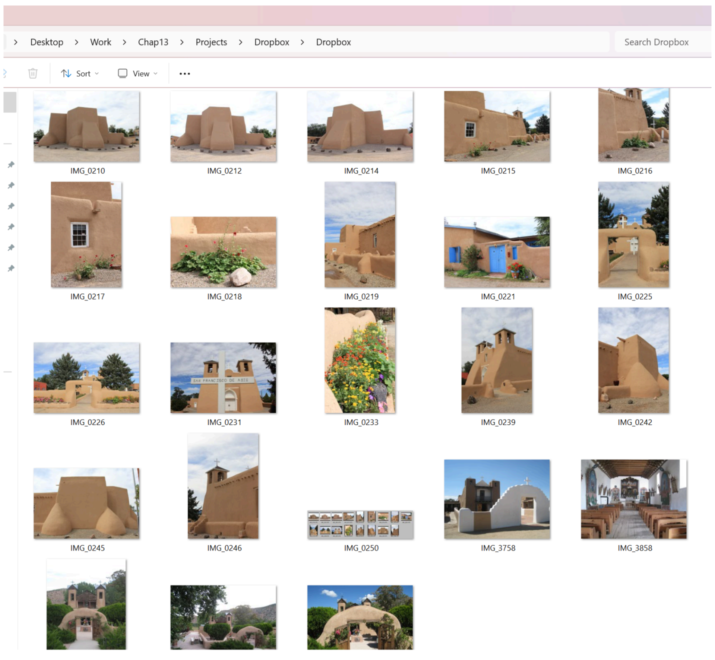
Exported the entire Dropbox directory structure locally to preserve evidence for further offline analysis.



9. Saving the Dropbox Folder: Exported the entire Dropbox directory structure locally to preserve evidence for further offline analysis.



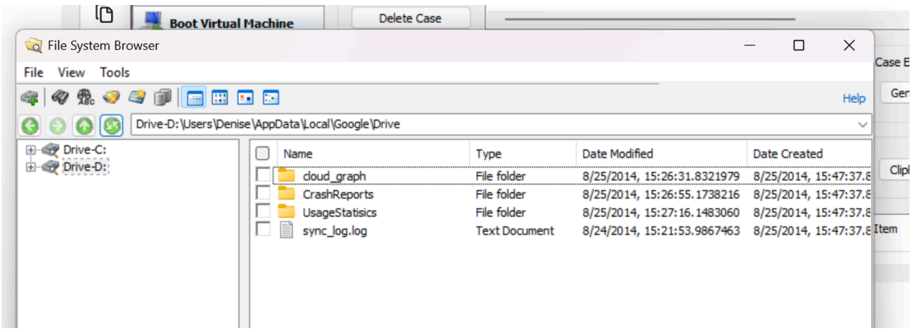
10. Reviewing Extracted Dropbox Content
Successfully extracted Dropbox contents to a working folder. Confirmed that the images from Dropbox.zip were accessible and viewable.



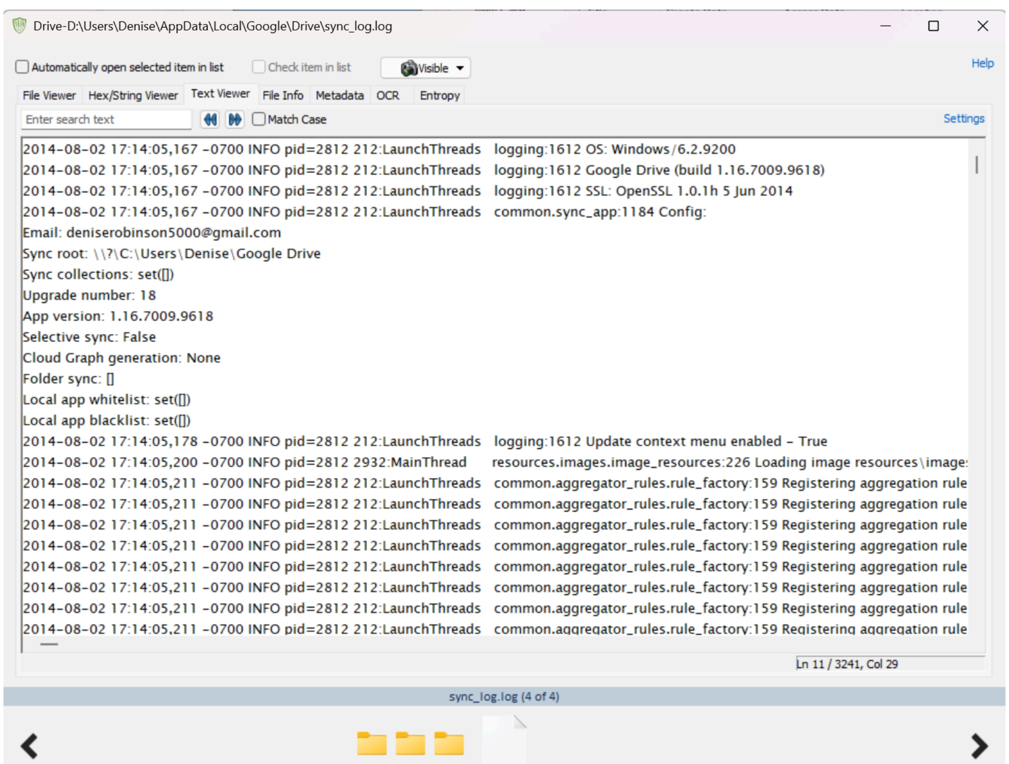
Summary for Hands-On Project 13-1:
I examined the Dropbox folder at D:\Users\Denise\Dropbox using OSForensics. It contained an empty .dropbox.cache folder, a .dropbox file, a desktop.ini file, a Dropbox.zip archive, and a Getting Started.pdf document. The PDF contained general cloud service information with no suspicious content. The main finding was Dropbox.zip, which included numerous JPEG and PNG images showing mainly architectural and scenic photos. The Dropbox folder was saved for further analysis, and no malicious artifacts were found.

Hands-On Project 13-2:

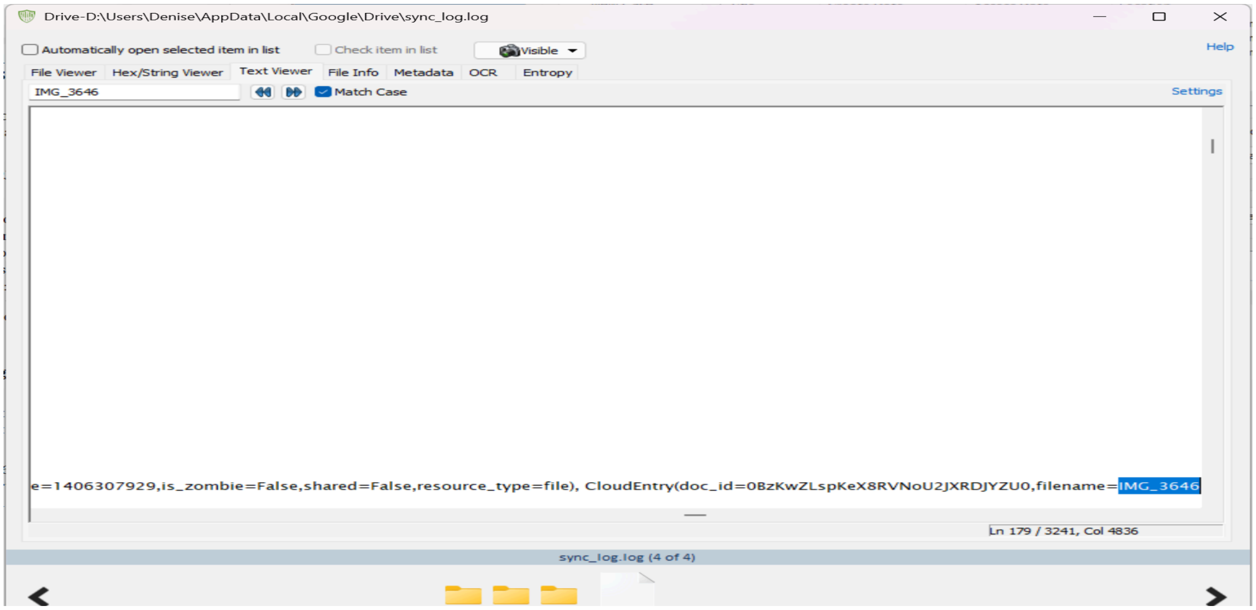
1. Google Drive Application Data Location:
Navigating to D:\Users\Denise\AppData\Local\Google\Drive reveals folders such as cloud_graph, CrashReports, UsageStatistics, and the sync_log.log file for the Google Drive application.



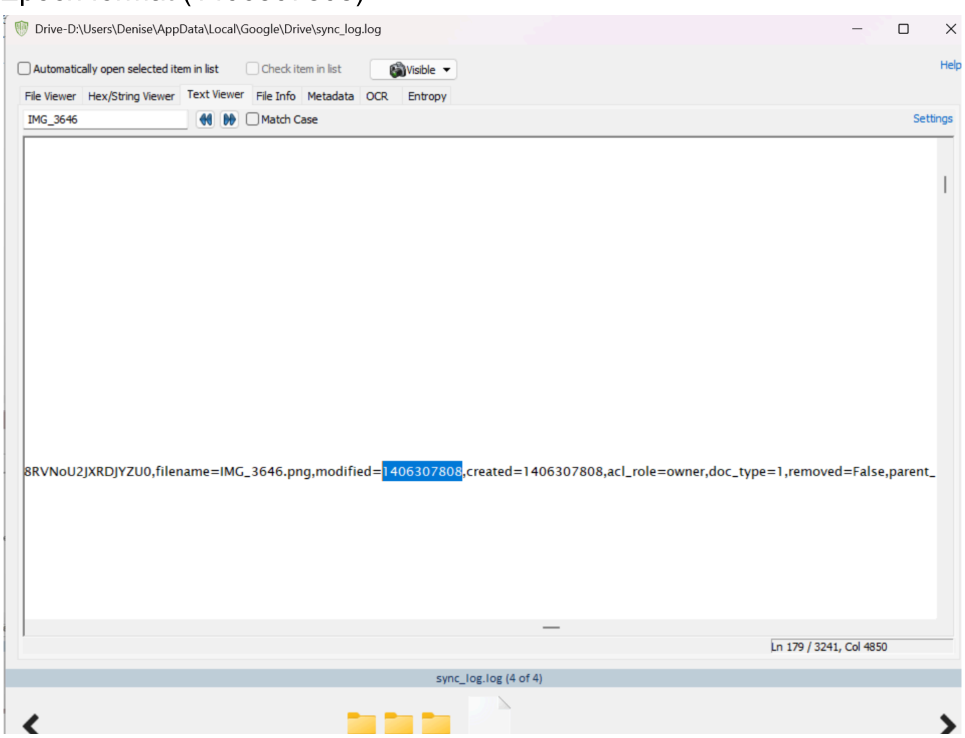
2. Viewing Google Drive sync_log.log:
Opening sync_log.log in OSForensics shows application version details (1.16.7009.9618) and logging information about Denise Robinson's Google Drive synchronization activities.



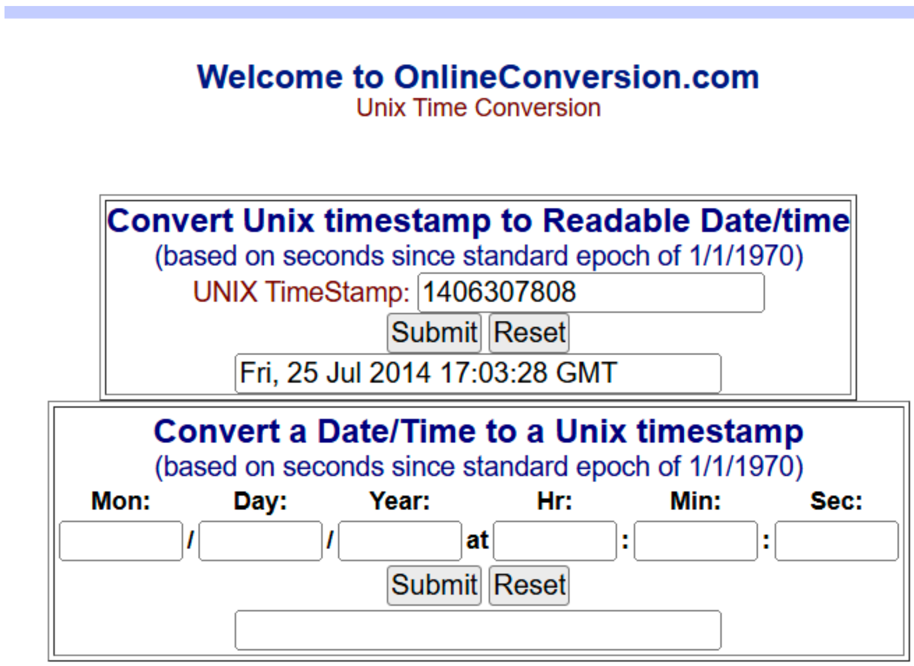
3. Searching for Specific File References:
A targeted search in sync_log.log for "IMG_3646" identifies a reference to a PNG image file that was part of the Google Drive sync operations.



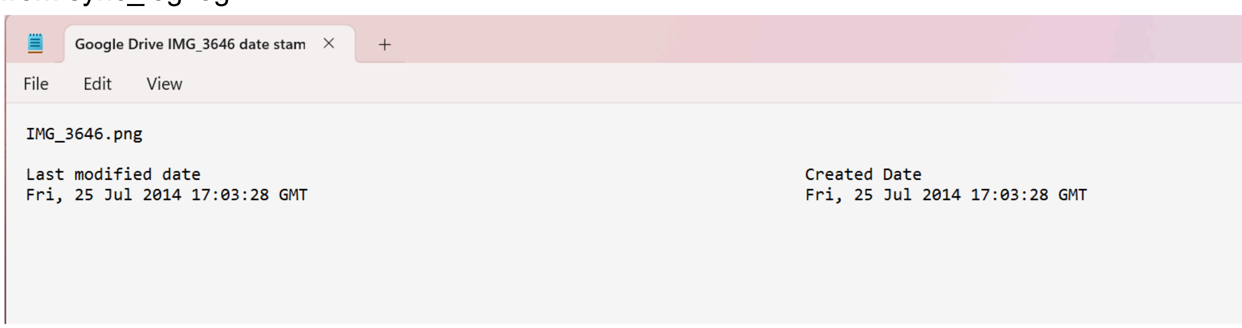
4. Modified Timestamp for IMG_3646:
Located the modified timestamp for IMG_3646.png within the sync_log.log file, recorded in Unix Epoch format (1406307808).



5. Converted Unix Timestamp to Human-Readable Date:
Using an online Unix time converter, the timestamp 1406307808 was converted to Fri, 25 Jul 2014 17:03:28 GMT.



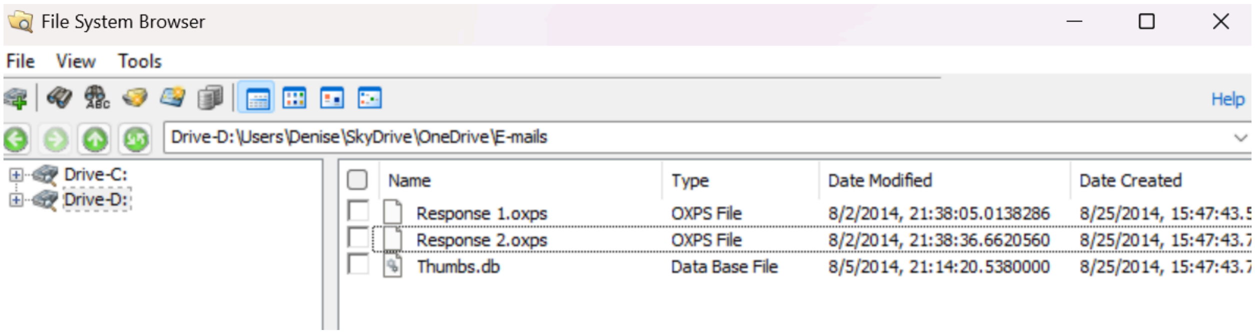
6. Documentation of IMG_3646 Timestamps:
The final document – Google Drive IMG_3646 date stamps.txt records both the creation and last modification dates for IMG_3646.png as Fri, 25 Jul 2014, matching the metadata extracted from sync_log.log.



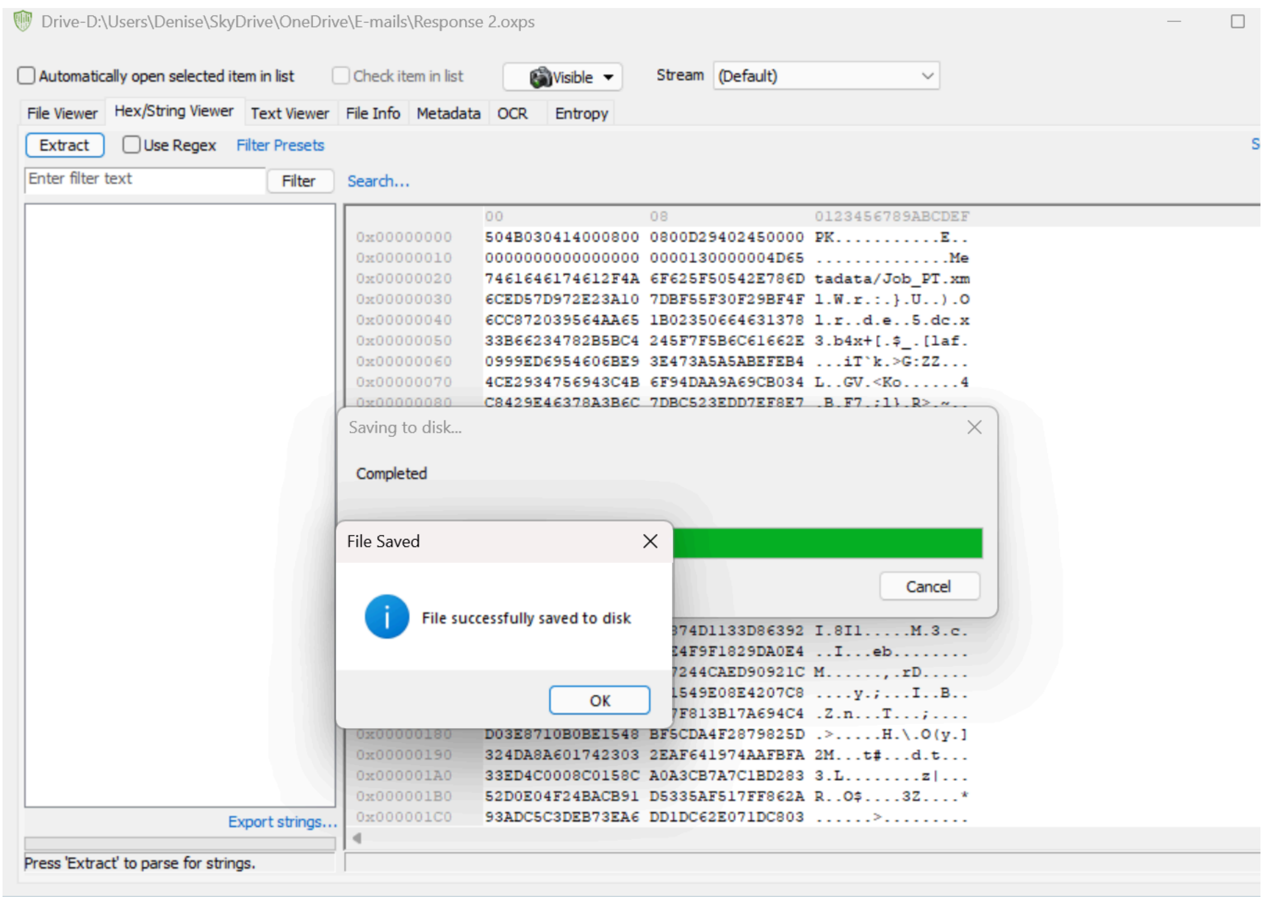
Summary for Hands-On Project 13-2:
In this project, I examined the Google Drive application data under D:\Users\Denise\AppData\Local\Google\Drive. I reviewed the sync_log.log file and found references to image files, including IMG_3646.png. Using the log data, I extracted Unix timestamps for the file's creation and modification, then converted them to human-readable format. The timestamps confirmed the file activity occurred on July 25, 2014. No suspicious modifications were detected beyond the standard sync operations.

Hands-On Project 13-3:

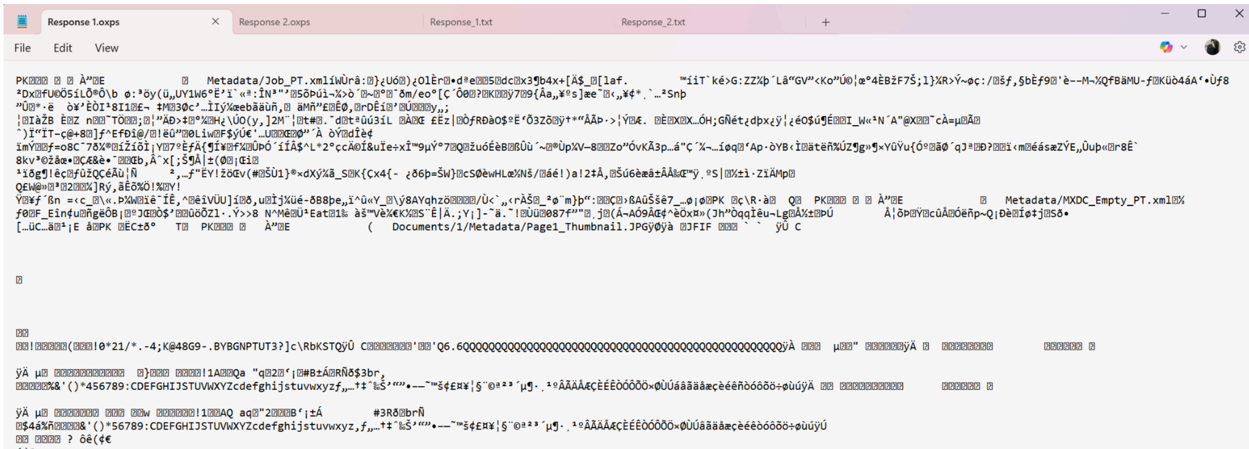
1. Located the OneDrive E-mails folder containing two .oxps files (Response 1 and Response 2) and a thumbs.db file, using OSForensics' File System Browser.



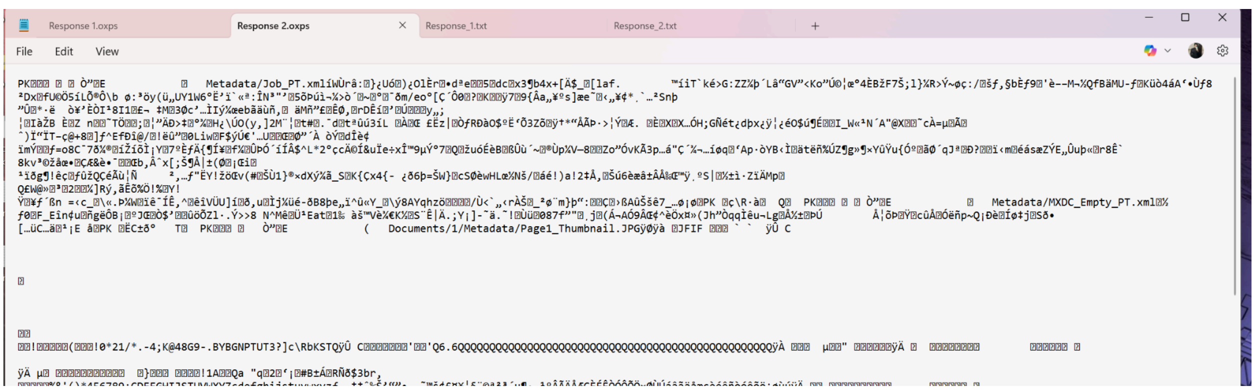
2. Extracted and saved Response 2.oxps from the forensic image to a local work folder for further analysis.



3. Opened Response 1.oxps directly; file contents appeared corrupted and unreadable, suggesting the need for string extraction.



4. Opened Response 2.oxps directly; file displayed as non-readable binary data, requiring additional processing to extract useful information.



5. Used OSForensics string extraction tool on Response 1.oxps; recovered metadata paths, image references, and document structure elements.

Response 1.oxps Response 2.oxps Response_1.txt Response_2.txt

File Edit View

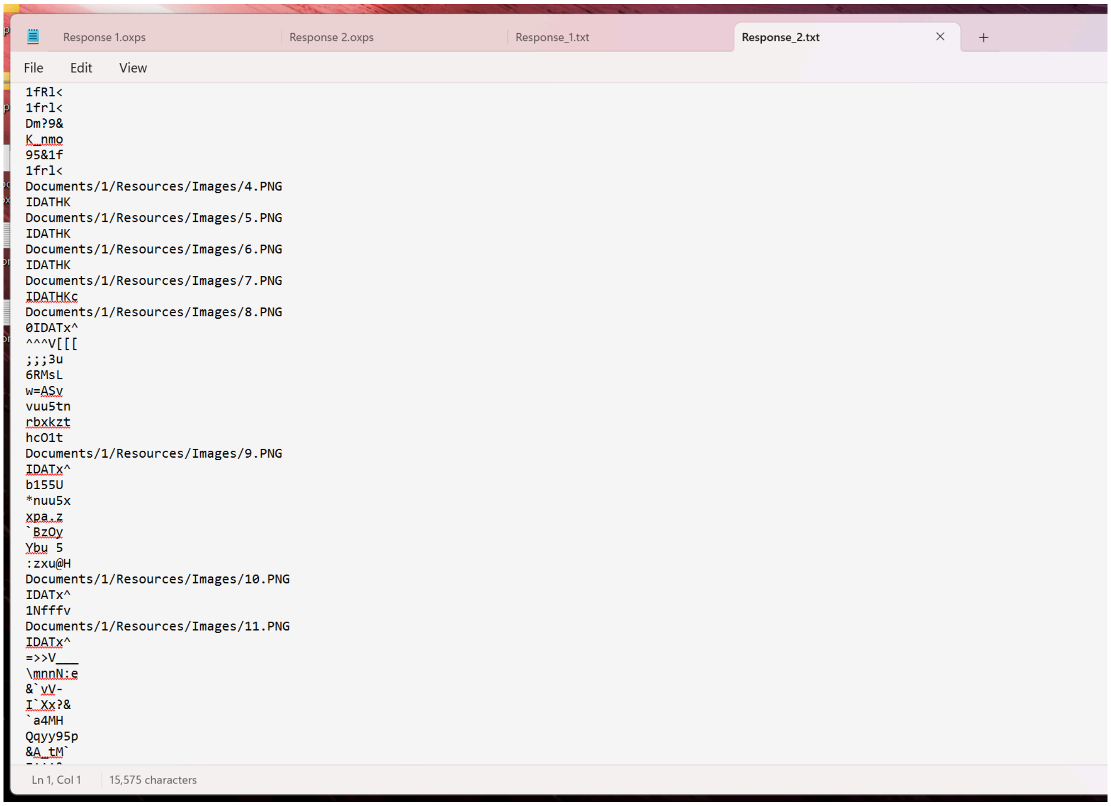
```
====================================================
Filepath:           Drive-D:\Users\Denise\SkyDrive\OneDrive\E-mails\Response 1.oxps
Min. String Length: 5
Max. String Length: 128
Repeat Char Limit:  5
Case Change Limit:  15
Special Characters: Yes
Number of strings:  1020
====================================================
-------------------Begin of List--------------------
Metadata/Job_PT.xml
b4x+[
[laf.
>G:ZZ
UY1W6
8AYqhz
Metadata/MXDC_Empty_PT.xml
Documents/1/Metadata/Page1_Thumbnail.JPG
0*21/*.-4;K@48G9-.BYBGNPTUT3?]c\RbKSTQ
*456789:CDEFGHIJSTUVWXYZcdefghijstuvwxyz
*56789:CDEFGHIJSTUVWXYZcdefghijstuvwxyz
YyW^x
-VtTH
Documents/1/Pages/1.fpage
Documents/1/Pages/_rels/1.fpage.relsm
Documents/1/Pages/2.fpage
yY^6u[
jQwuK
MXBK/
_0YBe
']W\.
Documents/1/Pages/_rels/2.fpage.rels
i0L*Q
Documents/1/Resources/Images/1.PNG
\IDATx^
B]hKqB
-0bG#
L[`pD]
pSD-8E
`7hG[
o]FQh
QL[4Oh
```

Response 1.oxps Response 2.oxps Response_1.txt Response_2.txt

File Edit View

```
Ti"SN
L5f8U
Documents/1/Pages/_rels/5.fpage.relsm
Documents/1/Pages/6.fpage
3pRM@c
t#K]4
Documents/1/Pages/_rels/6.fpage.relsm
Documents/1/Pages/7.fpage
*RqT`I
RAH.bh
B[wjW
;bZO?:
v77G/
Documents/1/Pages/_rels/7.fpage.relsm
Documents/1/Pages/8.fpage
Documents/1/Pages/_rels/8.fpage.relsm
Documents/1/Pages/9.fpage
Documents/1/Pages/_rels/9.fpage.relsm
Documents/1/Pages/10.fpage
Documents/1/Pages/_rels/10.fpage.relsm
Documents/1/Pages/11.fpage
Documents/1/Pages/_rels/11.fpage.relsm
Documents/1/Pages/12.fpage
Documents/1/Pages/_rels/12.fpage.relsm
Documents/1/Pages/13.fpage
Documents/1/Pages/_rels/13.fpage.relsm
Documents/1/Pages/14.fpage
Documents/1/Pages/_rels/14.fpage.relsm
Documents/1/Pages/15.fpage
Documents/1/Pages/_rels/15.fpage.relsm
Documents/1/Pages/16.fpage
Documents/1/Pages/_rels/16.fpage.relsm
Documents/1/Pages/17.fpage
]#a`8
H0Hvhy
`1qb19
yx9De
Documents/1/Pages/_rels/17.fpage.relsm
Documents/1/Resources/Fonts/40735858-7816-4246-8030-7B7F6B26ACEB.odttf
yS.ip4K
/X01@W
A:q\H
=hq8k5:
```

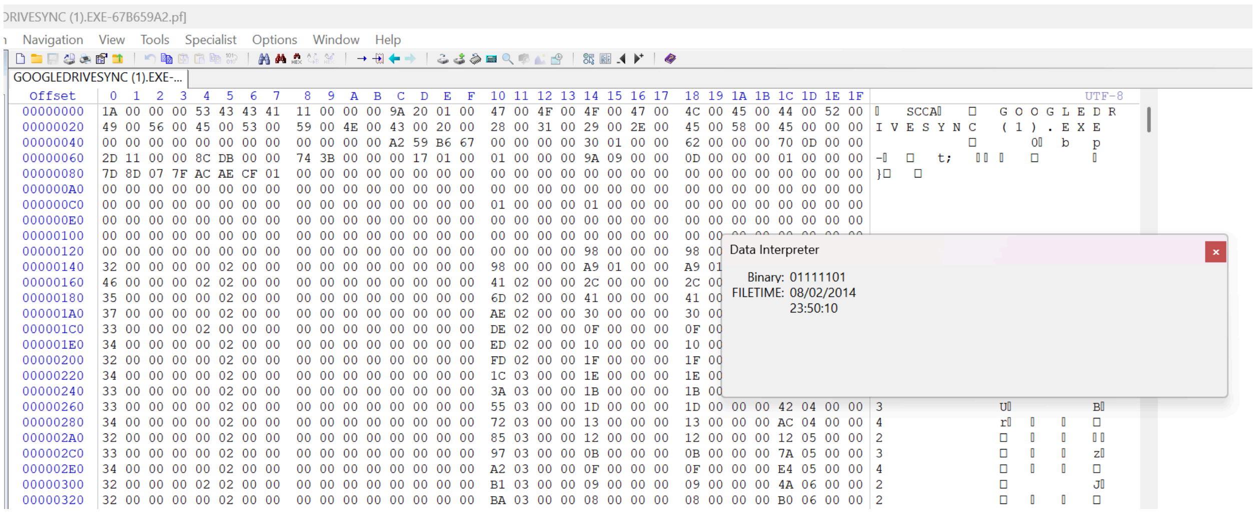
6. Extracted strings from Response 2.oxps; found references to PNG images, document page files, and metadata files.



Summary for Hands-On Project 13-3:
In this project, I examined the OneDrive synchronization folder for Denise Robinson to search for any email correspondence. I located two .oxps files, Response 1.oxps and Response 2.oxps, within the E-mails folder. When opened directly, both files appeared unreadable due to corrupted or compressed content. Using OSForensics, I extracted readable strings from both files, revealing metadata paths, references to image files, page resources, and document structures. While no direct readable email content was found, the extracted data indicated the presence of document components typically associated with structured files. All findings were saved for documentation and further review.

Hands-On Project 13-4:

Successfully opened the prefetch file GOOGLEDRIVESYNC(1).EXE-67B59A2.pf in WinHex for analysis. Used the Data Interpreter tool to find the last run date and time: 08/02/2014 at 23:50:10:



Verified the number of times the application was run as 52 times based on the value at offset 0xD4:

Memo for GOOGLEDRIVESYNC(1).EXE-67B59A2.pf:

Prefetch File Analyzed: GOOGLEDRIVESYNC(1).EXE-67B59A2.pf

Last Access Date and Time:
August 2, 2014 at 11:50:10 PM (23:50:10 GMT)

Number of Times Program Run:
52 times

Conclusion:
The analysis of the prefetch file confirmed that Denise Robinson last accessed Google Drive on August 2, 2014, and the Google Drive Sync application had been executed a total of 52 times.